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April, 2026

Cross-Country Differences in Used Car Depreciation: Age vs. Mileage Effects in Germany and the Netherlands

1. Introduction

The pricing of used cars is an important factor for both buyers and sellers in secondary vehicle markets. Vehicle characteristics such as age, mileage, and transmission type are widely recognized as key determinants of resale value, as they directly reflect a car's condition and expected future performance. Understanding how these factors influence prices is essential for market participants seeking to make informed decisions.

While these determinants are generally well understood, less attention has been paid to how their effects may differ across countries. Differences in consumer preferences, market structures, and regulatory environments could lead to variation in how vehicle characteristics are valued. In particular, this raises the question of whether depreciation patterns differ between national markets.

This paper examines how mileage, age, and transmission type affect used car prices, and whether cars depreciate differently in **Germany** compared to the **Netherlands**. Adopting a marketplace perspective, the aim is to provide **insights that are directly relevant for sellers** operating in these markets.

2. Data

2.1 Data collection

Data was scraped from Mobile.de using the Apify actor “Mobile.de Scraper” (<https://console.apify.com/actors/5rqwyghNIbO6VMnlX/input>), created by 3x1t. This automated tool extracted used car listings from Germany and the Netherlands on 24 April 2026. Multiple scraping runs ensured a balanced sample across fuel types (petrol, diesel, electric, hybrid, plug-in hybrid) and vehicle categories, yielding a final dataset of 4,862 observations: 2,441 from Germany and 2,421 from the Netherlands.

All data processing, cleaning, and analysis was conducted in Python using Jupyter notebooks in VS Code. This workflow handled data harmonization, outlier removal, variable construction (log transformations, mean-centering), and OLS regression estimation.

2.2 Data cleaning

Datasets from both countries were harmonized to include consistent variables: listing price, mileage (km), year of production (to derive age), transmission type, fuel type, and collection date. Filtering criteria were applied to remove outliers:

- Price: 3,000–50,000 EUR
- Mileage: < 220,000 km
- Age: 1–15 years
- Implausible cases: vehicles >10 years old with <50,000 km mileage

Fuel type categories were standardized across countries (e.g., “Petrol, E10-enabled” → petrol; “Diesel, Biodiesel Suitable” → diesel). Some categories were combined to ensure comparability and sufficient observations per group.

2.3 Variable construction

Key transformations improved model performance and interpretability:

Table 1: Constructed variables

Original	Constructed	Purpose
Price	$\log(\text{Price})$	Linearize, % interpretation
Mileage (or km)	$\log(\text{km})$	Elasticity interpretation
Age	c_{Age}	Interaction interpretability

Controls and interactions:

- is_manual : 1 = manual transmission
- $Fuel$ dummies: electric, diesel, hybrid, plug-in hybrid (petrol omitted)
- is_NL : 1 = Netherlands
- Interactions: $c_{age} \times is_manual$, $c_{age} \times is_NL$, $c_{log_km} \times is_NL$

2.4 Descriptive overview

Table 2: Key Descriptive Statistics by Country, Averages

Variable	Germany	Netherlands
Price (EUR)	20,631	20,953
Mileage (km)	92,353	125,966
Age (years)	8.20	6.63
$\log(\text{Price})$	9.90	9.82
$\log(\text{km})$	11.41	11.65

Fuel type distribution is identical across countries by construction.

Price correlations

$$cor_{price, age} = -0.269$$

$$cor_{price, mileage} = -0.293$$

3. Methodology

3.1 Data source and sample

Data was collected from Mobile.de, covering 4,862 used car listings from Germany (2,441) and the Netherlands (2,421). Multiple web scraping rounds ensured balance across fuel

types. The sample was restricted to vehicles aged 1-15 years, priced 3,000-50,000 EUR, and with mileage below 220,000 km. Implausible outliers were removed.

3.2 Variables

The dependent variable is log price ($\log[Price]$). Main explanatory variables are mean-centered age (c_age) and logged mileage (c_log_km). Controls include transmission type ($is_manual = 1$), fuel type dummies (electric, diesel, hybrid, plug-in hybrid; petrol omitted, only statistically significant fuel dummies were kept in the model), and country ($is_NL = 1$ for Netherlands). Interaction terms test whether age and mileage effects differ by country and transmission.

3.3 Model specification

OLS Model

$$\begin{aligned} \log(price_i) = & \beta_0 + \beta_1 is_manual_i + \beta_2 c_age_i + \beta_3 [c_age_i \times is_manual_i] + \\ & + \beta_4 c_log_km_i + \beta_5 is_electric_i + \beta_6 is_diesel_i + \beta_7 is_NL_i + \\ & + \beta_8 [c_age_i \times is_NL_i] + \beta_9 [c_log_km_i \times is_NL_i] + \varepsilon_i \end{aligned}$$

The baseline is a petrol, automatic car in Germany. All coefficients are relative to this reference.

Table 3: units of measure

variable	unit	variable	unit
price	EUR	is_electric	boolean
is_manual	boolean	is_diesel	boolean
age	years	is_NL	boolean
km	kilometer		

4. Results

4.1 Model fit

The OLS regression explains 41.5% of the variation in log car prices ($R^2 = 0.415$, adjusted $R^2 = 0.414$). The model is highly significant overall (F-statistic, $p < 0.001$), confirming that the included variables jointly predict listing prices.

4.2 Transmission and fuel effects

Manual transmission cars trade at a substantial discount relative to automatic cars. The interaction term shows manual cars depreciate slightly more slowly with age, though this effect is small in magnitude. Fuel type controls are significant: electric cars trade at an 11.1% discount relative to petrol, while diesels command a 9.7% premium.

4.3 Country price levels

Dutch listings are 6.4% cheaper than comparable German listings (*is_NL coefficient* = $\beta_7 = 0.0666$, $p < 0.05$). This reflects baseline market price differences unrelated to vehicle characteristics.

4.4 Depreciation patterns by country

Age effects are similar across markets. In Germany, each additional year reduces prices by 2.8%. In the Netherlands, the effect is slightly larger at 3.1%, but the country interaction is insignificant ($p = 0.373$).

Mileage effects differ sharply. The German mileage elasticity implies a 0.83% price drop per 10% mileage increase. In the Netherlands, this rises to 2.9%. The country-mileage interaction is highly significant ($p < 0.001$).

4.5 Summary table

Table 4: Price depreciation effect of Age and Mileage

Country	Additional year of Age	Additional 10% increase of Mileage
Germany	-2,8%	-0,083%
Netherlands	-3,1%	-2,910%

The full OLS regression model is in the appendix.

4.6 Main finding

Depreciation patterns differ between markets, driven primarily by mileage valuation. Age effects are comparable, but Dutch buyers penalize high-mileage cars much more heavily than German buyers. This constitutes the paper's central result.

5. Discussion

5.1 Main findings

The analysis reveals that used car depreciation patterns differ between Germany and the Netherlands, but not uniformly. Age-related depreciation is nearly identical across both markets (~3% per year). However, mileage carries a much heavier penalty in the

Netherlands (2.9% vs 0.83% price drop per 10% mileage increase), representing the paper's central result.

5.2 Economic interpretation

Consumer preferences: Dutch buyers appear more sensitive to mileage as a signal of wear, reliability, and future maintenance costs. German buyers seem more tolerant of usage intensity, possibly valuing other attributes (brand, condition) more heavily.

Market structure: Germany's larger, more liquid used car market offers buyers greater choice, potentially reducing mileage sensitivity. The smaller Dutch market may amplify price adjustments based on observable usage.

Institutional factors: Higher Dutch vehicle usage costs (taxes, fuel) likely increase mileage's role as a total-cost indicator, explaining its outsized pricing impact.

5.3 Implications

For sellers: High-mileage cars fetch relatively better prices in Germany; low-mileage cars command premiums in the Netherlands. Cross-border valuation differences suggest potential arbitrage opportunities, though transaction costs likely limit this.

For research: The findings demonstrate that depreciation is not purely mechanical but shaped by market-specific valuation rules. Mileage functions less as objective wear and more as a market signal that varies by context.

5.4 Limitations

The analysis omits brand, model, and condition data, creating potential omitted variable bias. Data from one platform (Mobile.de) may not represent full market dynamics, particularly private sales. The cross-sectional design captures pricing at a single point (April 2026) rather than true depreciation trajectories.

6. Conclusion

This study demonstrates that used car depreciation patterns differ between Germany and the Netherlands, primarily through mileage valuation rather than age effects. Using OLS regression on 4,862 Mobile.de listings, the analysis found age reduces prices similarly across both markets (~3% per year). However, mileage carries a much heavier penalty in the Netherlands (2.9% vs. 0.83% price drop per 10% mileage increase), representing the paper's central result.

The findings reveal that depreciation reflects market-specific preferences beyond mechanical wear. Dutch buyers treat mileage as a stronger signal of total ownership cost, while German buyers appear more tolerant of usage intensity. This suggests potential cross-border valuation differences for sellers, though transaction costs likely limit arbitrage opportunities.

Future research should incorporate brand/model data, transaction prices rather than listings, and longitudinal designs tracking individual vehicles over time. Extending the analysis to additional European markets would test the generalizability of these national valuation patterns.

7. Appendix

7.1 OLS Regression Summary

Dep. Variable:	log_price	R-squared:	0.415
Model:	OLS	Adj. R-squared:	0.414
Method:	Least Squares	F-statistic:	382.5
Date:	Sat, 04 Apr 2026	Prob (F-statistic):	0.00
Time:	21:49:11	Log-Likelihood:	-1421.0
No. Observations:	4862	AIC:	2862.
Df Residuals:	4852	BIC:	2927.
Df Model:		9	
Covariance Type:		nonrobust	

	coef	std err	t	P> t	[0.025	0.975]
const	10.0203	0.009	1145.487	0.000	10.003	10.037
is_manual	-0.5045	0.012	-43.569	0.000	-0.527	-0.482
c_age	-0.0280	0.003	-10.178	0.000	-0.033	-0.023
c_age__isManual	0.0084	0.004	2.201	0.028	0.001	0.016
c_log_km	-0.0828	0.031	-2.688	0.007	-0.143	-0.022
isElectric	-0.1179	0.018	-6.550	0.000	-0.153	-0.083
isDiesel	0.0927	0.015	6.095	0.000	0.063	0.123
is_NL	-0.0666	0.011	-6.137	0.000	-0.088	-0.045
c_age__NL	-0.0032	0.004	-0.892	0.373	-0.010	0.004
c_log_km__NL	-0.2077	0.034	-6.054	0.000	-0.275	-0.140

Omnibus:	86.574	Durbin-Watson:	1.934
Prob(Omnibus):	0.000	Jarque-Bera (JB):	162.639
Skew:	0.095	Prob(JB):	4.82 e-36
Kurtosis	3.876	Cond. No.	34.6